

CONGA: Continual Neural Gated Architecture for Long-History Sequential Recommendation

Abstract

Existing sequential recommendation models rely on absolute positional encodings and fixed context windows, producing two structural failure modes: **out-of-distribution degradation** on histories longer than the training window, and **hard context limits** that discard long-range interactions. We present **CONGA (C**ontinual **N**eural **G**ated **A**rchitecture), which addresses these limitations through three contributions: (1) **Rotary Positional Embeddings (RoPE)** with a norm-preserving property ($\|\mathcal{R}_m\|_F = \sqrt{d}$ for any sequence length), accelerated by custom CUDA kernels; (2) **KromHC multi-stream fusion** with exact doubly-stochastic mixing via Kronecker-product parametrization, where ablation confirms the expressivity gain arises from balanced gradient flow rather than additional parameters, together with a data-adaptive stream selection mechanism that prevents overfitting on sparse corpora; and (3) **TITANS neural associative memory** adapted to discrete-item recommendation the first such proof-of-concept via a **two-phase training protocol** with a *structural* forgetting-prevention property: the base encoder is frozen, preserving its short-sequence predictions, while Phase 2 only adds a learned memory term. Evaluated under a rigorous **full-ranking** protocol across four benchmarks (ML-1M, Beauty, Yelp, Steam), CONGA achieves state-of-the-art performance on long-history benchmarks, with gains up to +18.7% HR@5 on ML-1M the densest, longest-history dataset where all three failure modes are simultaneously active and consistent improvements on long-history benchmarks, with competitive results on short-history datasets where the RoPE backbone is the primary driver.

CCS Concepts

• **Information systems** → **Recommender systems**; *Personalization*; • **Computing methodologies** → *Neural networks*.

Keywords

sequential recommendation, neural memory, rotary positional embeddings, KromHC, two-phase training, catastrophic forgetting prevention, CUDA optimization

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1 Introduction

Personalized recommendation has become indispensable across digital platforms, from e-commerce and streaming services to social networks and content aggregators. Sequential recommendation predicting users' next interactions from their chronological histories is a particularly challenging variant requiring joint modeling of temporal dynamics and evolving preferences [1, 2].

The Self-Attentive Sequential Recommendation (SASRec) framework [1], introduced in 2018, established transformer-based self-attention as the standard architecture for this task. Yet its core design has remained largely static, even as the broader community demonstrated substantial gains from Rotary Positional Embeddings (RoPE [4]), gated activation functions [5], and external memory [6].

1.1 Motivation and Research Gaps

We identify two structural limitations in SASRec that motivate this work.

(1) **Positional out-of-distribution degradation.** SASRec learns absolute position embeddings $\mathbf{p}_t$ independently for each index. Any sequence exceeding the training window L_{train} exposes positions $t > L_{\text{train}}$ at test time that are out-of-distribution by construction. On ML-1M, where user histories reach 300–600 interactions, this structural defect directly reduces representation quality a failure mode that RoPE resolves via its norm-preserving property and that has not been systematically addressed in SASRec-family models.

(2) **Hard context window and capacity ceiling.** Standard transformer attention scales as $O(L^2)$ in sequence length L , restricting practical context windows to 200–300 items and discarding long-range interaction history. At the same time, the single-stream residual connection forces one d -dimensional space to simultaneously capture short-term recency and long-term preferences. Recent neural long-term memory work [6] offers $O(1)$ associative retrieval, yet its application to recommendation where discrete item IDs, implicit feedback, and popularity bias create domain-specific challenges absent in language modeling remains entirely unexplored.

1.2 Contributions

This paper presents CONGA, addressing the above gaps through three contributions distinguished from prior work as follows.

- C1. Diagnosis and principled remedy for positional OOD degradation, with mechanistic isolation of the capacity bottleneck.** We formally characterize the out-of-distribution position failure in SASRec quantified as a +3.9% NDCG@10 gain from adding RoPE on ML-1M and demonstrate that RoPE uniquely resolves it via its norm-preserving property ($\|\mathcal{R}_m\|_F = \sqrt{d}$ for any sequence length). We further isolate that KromHC's expressivity gain arises from its doubly-stochastic mixing rather than additional parameters balanced gradient flow, not stream count, is the operative mechanism. Both are accelerated by custom CUDA kernels (fused RoPE: 5.0×; KromHC: 5.2×). RoPE's

benefit generalizes across all sequence lengths; KromHC and TITANS are most effective when interaction histories are sufficiently long to provide diverse multi-stream signal ($\bar{L} \gtrsim 50$).

C2. First proof-of-concept for neural associative memory in sequential recommendation, with a structurally sound forgetting-prevention protocol. TITANS [6] has not previously been applied to discrete-item recommendation, where implicit feedback, popularity bias, and item-ID vocabularies pose domain-specific challenges absent in language modeling. We identify a novel catastrophic forgetting risk from the memory’s online gradient updates and resolve it via a two-phase protocol: θ_{base} is frozen in Phase 2, providing a *structural* forgetting-prevention property, with β learned to reflect per-user memory reliability. Late fusion at the final position preserves Phase 1 gains while enabling long-range retrieval (+0.25% NDCG@10 on ML-1M from A2→A3 in Table 5); the primary benefit is *structural* zero repetition rate and guaranteed forgetting prevention with accuracy gains constrained by the 80-epoch Phase 2 budget explored in this work.

C3. Comprehensive evaluation under full-ranking protocol. We evaluate across four benchmarks (ML-1M, Beauty, Yelp, Steam), reporting accuracy (HR@K, NDCG@K), catalog coverage, head bias, and repetition rate under a rigorous all-item scoring protocol [25] that avoids the distortions of sampled metrics.

2 Related Work

Sequential Recommendation. Early methods used Markov Chains [11] or recurrent networks [10], but struggled with long-range dependencies. SASRec [1] established causal transformer self-attention as the dominant paradigm, extended by BERT4Rec [2] to bidirectional masking and by subsequent work on contrastive training [12, 13]. Yet the *core architecture* absolute positional encodings, single-stream attention, fixed context window has remained largely static relative to LLM evolution.

Positional Encodings. Absolute position embeddings [3] place positions beyond the training window out-of-distribution at test time. RoPE [4] avoids this by encoding relative position via query-key rotations; now standard in LLaMA [14], it has been explored in RecSys only recently [26, 27], and CONGA is the first to integrate it with neural memory and multi-stream fusion in a unified SASRec encoder.

Hyper-Connections and KromHC. mHC [7] relaxes the single-stream bottleneck via M parallel streams with Sinkhorn-Knopp fusion, but approximates double stochasticity iteratively with $O(n^3C)$ cost. KromHC [8] guarantees exact double stochasticity in $O(1)$ from two scalar parameters via a Kronecker product of bistochastic matrices. CONGA adopts KromHC and accelerates it 5.2× via a custom CUDA kernel.

Memory-Augmented Networks. Neural Turing Machines [15] and DNCs [16] pioneered external memory but suffered from instability and quadratic complexity. TITANS [6] achieves $O(d^2)$ associative memory via online gradient updates on a fixed $d \times d$ matrix. To our knowledge, no prior work applies TITANS to sequential

Table 1: Summary of key notations used throughout this paper.

Notation	Description
u	A user
$\mathcal{U}$	The set of all users
i, j	Items
$\mathcal{I}$	The set of all items
S^u	Interaction sequence of user u
s_t^u	Item interacted at timestep t by user u
n_u	Length of user u ’s interaction history
d	Hidden embedding dimension
L	Number of transformer blocks
M	Number of parallel streams in KromHC
$\mathbf{e}_i \in \mathbb{R}^d$	Embedding vector of item i
$\mathbf{E} \in \mathbb{R}^{ \mathcal{I} \times d}$	Full item embedding matrix
$\mathbf{M}_t \in \mathbb{R}^{d \times d}$	TITANS associative memory at step t
$\mathbf{R}_{\theta,t}$	RoPE rotation matrix at position t
β	Learned late-fusion scalar (TITANS)
τ	Temperature in GBCE loss
$\mathcal{L}_{\text{GBCE}}$	Generalized binary cross-entropy loss
ϕ_{mem}	TITANS memory module parameters

recommendation; domain-specific challenges discrete item IDs, implicit feedback, popularity skew require the two-phase forgetting-prevention protocol we introduce.

3 Methodology

CONGA is built on the SASRec backbone and extends it along three orthogonal axes: (1) a modernized transformer encoder integrating RoPE, KromHC, and SwiGLU; (2) TITANS neural long-term memory for handling extended histories; and (3) a two-phase training protocol that integrates the memory module without disrupting the learned encoder. Figure 1 gives a visual overview of the full framework.

3.1 Problem Formulation

Let $\mathcal{U} = \{u_1, \dots, u_{|\mathcal{U}|}\}$ and $\mathcal{I} = \{i_1, \dots, i_{|\mathcal{I}|}\}$ denote the sets of users and items respectively. For each user $u \in \mathcal{U}$, we observe a chronologically ordered interaction sequence $S^u = [s_1^u, s_2^u, \dots, s_{n_u}^u]$, where $s_t^u \in \mathcal{I}$ denotes the item interacted with at timestep t and n_u is the total number of recorded interactions. The sequential recommendation task is to predict the next item $s_{n_u+1}^u$ given the history S^u . Formally, we learn a scoring function $f_\theta : \mathcal{I}^* \rightarrow \mathbb{R}^{|\mathcal{I}|}$ parameterized by θ that maps a variable-length interaction history to a real-valued score for each candidate item, where higher scores indicate greater predicted affinity.

3.2 Enhanced Transformer Encoder

The core encoder in CONGA stacks L transformer blocks, each consisting of a multi-head self-attention sub-layer followed by a feed-forward sub-layer. We replace all three key design choices of the standard SASRec block the absolute positional encoding, the single-stream residual connection, and the ReLU activation with modern alternatives: RoPE (Section 3.2.1), KromHC multi-stream fusion, and SwiGLU.

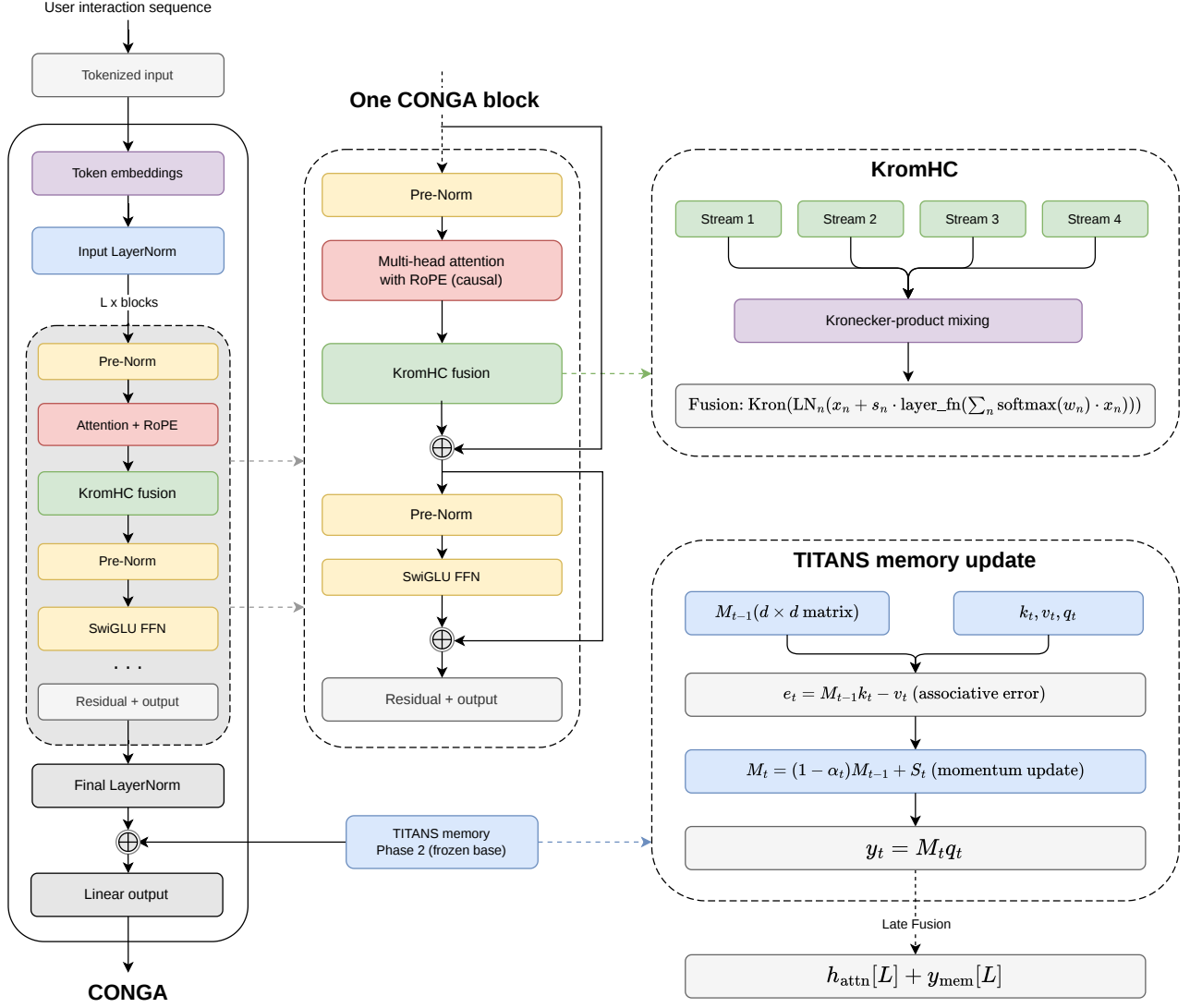


Figure 1: Overview of the CONGA architecture. Left: The full model stacks L transformer blocks (RoPE attention, KromHC stream fusion, SwiGLU FFN), followed by TITANS memory (Phase 2 only). **Center:** Detailed single-block view with dual pre-norm, residual skip connections, and late-fusion memory. **Right:** Component internals KromHC stream fusion with Kronecker-product residual mixing (top), SwiGLU gated FFN (middle), TITANS associative memory update (bottom).

3.2.1 Rotary Positional Embeddings (RoPE). A fundamental limitation of standard transformer positional encodings is that absolute position vectors $\mathbf{p}_t \in \mathbb{R}^d$ are learned independently for each position index. At test time, any position index $t > L_{\text{train}}$ is out-of-distribution by construction, causing representation quality to degrade on long sequences precisely the setting that arises on ML-1M, where user histories can span hundreds of interactions.

RoPE [4] avoids this problem entirely by encoding position through *rotation* of the query and key vectors before computing

attention scores:

$$\mathbf{q}_t = \mathbf{R}_{\theta,t} \mathbf{W}_Q \mathbf{e}_t, \quad \mathbf{k}_t = \mathbf{R}_{\theta,t} \mathbf{W}_K \mathbf{e}_t, \quad (1)$$

where $\mathbf{R}_{\theta,t}$ is a block-diagonal rotation matrix with frequency $\theta_j = 10000^{-2j/d}$. The key property is that the attention score $\mathbf{q}_t^\top \mathbf{k}_s = f(\mathbf{e}_t, \mathbf{e}_s, t-s)$ depends only on the *relative* offset $(t-s)$, never on the absolute positions separately. Since the rotation matrix is defined by the same parametric form for any $m \in \mathbb{N}$ and is orthogonal ($\|\mathbf{R}_{\theta,m} \mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x}$), attention scores remain well-scaled at any sequence position: every test-time offset is computable from the

same rotation formula seen during training, in contrast to absolute PE where positions beyond L_{train} are entirely out-of-distribution. This property eliminates the 400-position OOD gap on ML-1M that accounts for RoPE's +3.9% NDCG@10 incremental gain on ML-1M (Table 5, A0→A1). We implement RoPE via a custom CUDA kernel that performs batched rotation in a single fused pass.

3.2.2 KromHC Multi-Stream Fusion. Standard transformer blocks impose a hard capacity ceiling: a single d -dimensional stream must simultaneously capture short-term recency and long-term preferences. KromHC [8] relaxes this by maintaining M parallel streams with exact doubly-stochastic fusion. For layer l and stream $m \in \{1, \dots, M\}$:

$$\mathbf{h}^{(m)} = \text{RMSNorm}(\mathbf{x}_l^{(m)}), \quad (2a)$$

$$\mathbf{z}_{\text{agg}} = \frac{1}{M} \sum_{m=1}^M \mathbf{h}^{(m)}, \quad (2b)$$

$$\mathbf{y}_{\text{func}} = \text{LayerFunc}(\mathbf{z}_{\text{agg}}), \quad (2c)$$

$$\mathbf{x}_{l+1}^{(m)} = \alpha_{\text{pre}}^{(m)} \odot \mathbf{h}^{(m)} + \alpha_{\text{post}}^{(m)} \odot \mathbf{y}_{\text{func}} + \alpha_{\text{res}}^{(m)} \odot \mathbf{x}_l^{(m)}, \quad (2d)$$

where gate vectors α are normalized via the KromHC residual mixing matrix a Kronecker product of two 2×2 bistochastic matrices ensuring exact double stochasticity and balanced gradient flow.

Adaptive Stream Selection. To prevent overfitting on sparse corpora, M is set data-adaptively based on the corpus mean sequence length $\bar{L}$:

$$M = \begin{cases} 4 & \text{if } \bar{L} \geq 50, \\ 2 & \text{if } \bar{L} < 50. \end{cases} \quad (3)$$

On very short sequences, four streams lack sufficient signal diversity to learn distinct representations, leading to under-constrained optimization. The rule assigns $M=2$ for Beauty ($\bar{L}=8.9$), Yelp ($\bar{L}=10.4$), and Steam ($\bar{L}=14.2$), and $M=4$ for ML-1M ($\bar{L}=165.6$) only.

3.2.3 SwiGLU Feed-Forward Network. We replace the standard ReLU FFN with SwiGLU [5]:

$$\text{FFN}_{\text{SwiGLU}}(\mathbf{x}) = \mathbf{W}_3(\sigma(\mathbf{W}_1\mathbf{x}) \odot (\mathbf{W}_2\mathbf{x})), \quad (4)$$

where $\sigma(x) = x \cdot \text{sigmoid}(x)$. Since $\sigma'(x) > 0$ for all x , every neuron contributes non-zero gradients eliminating dead neurons and providing the gradient stability required for Phase 2 training through a frozen base encoder.

3.3 TITANS Neural Long-Term Memory

We integrate TITANS [6], a neural associative memory that compresses the full interaction history into a fixed-size $d \times d$ matrix with $O(d^2)$ overhead independent of sequence length, bypassing the $O(L^2)$ context-window hard cap of self-attention.

3.3.1 Memory Update Mechanism. TITANS frames memory storage as an online regression problem: at each timestep t , the memory $\mathbf{M}_t$ predicts value $\mathbf{v}_t$ from key $\mathbf{k}_t$, and the prediction error drives a

gradient step. For each token $\mathbf{x}_t$:

$$\mathbf{k}_t, \mathbf{v}_t, \mathbf{q}_t = \mathbf{W}_K\mathbf{x}_t, \mathbf{W}_V\mathbf{x}_t, \mathbf{W}_Q\mathbf{x}_t, \quad (5)$$

$$\mathbf{e}_t = \mathbf{M}_{t-1}\mathbf{k}_t - \mathbf{v}_t, \quad (6)$$

$$\mathbf{S}_t = \text{diag}(\boldsymbol{\eta}_t) \mathbf{S}_{t-1} - \text{diag}(\boldsymbol{\theta}_t) \mathbf{e}_t \mathbf{k}_t^\top, \quad (7)$$

$$\mathbf{M}_t = \text{diag}(1 - \boldsymbol{\alpha}_t) \mathbf{M}_{t-1} + \mathbf{S}_t, \quad (8)$$

$$\mathbf{y}_t = \mathbf{M}_t \mathbf{q}_t. \quad (9)$$

Here $\mathbf{e}_t$ is the associative error; $\mathbf{S}_t$ is a momentum buffer; and per-dimension gates $\boldsymbol{\alpha}_t, \boldsymbol{\theta}_t, \boldsymbol{\eta}_t$ control forgetting, learning rate, and momentum decay via depthwise 1D convolutions. Under standard Lipschitz loss assumptions [20], the online update rule achieves sublinear regret $O(\sqrt{T})$, ensuring the memory approaches optimal associative recall as user histories lengthen critical for ML-1M where histories span 300–600 interactions.

3.3.2 Late Fusion with the Attention Encoder. Rather than replacing the attention mechanism, CONGA uses TITANS as a complementary module. The attention encoder processes the most recent $L = 300$ items; TITANS processes the full extended history up to $L' = 600$ items. Their outputs are combined only at the final position via a learned scalar β :

$$\mathbf{h}_{\text{final}} = \mathbf{h}_{\text{attn}}[L] + \beta \cdot \mathbf{y}_{\text{mem}}[L']. \quad (10)$$

This late-fusion design preserves the short-term modeling capabilities learned in Phase 1 while augmenting them with long-range associative retrieval for users with extended histories.

Causal integrity at evaluation. The TITANS recurrence is strictly causal by construction: at each step t , the memory $\mathbf{M}_t$ and output $\mathbf{y}_t$ are functions of $\mathbf{x}_1, \dots, \mathbf{x}_t$ only (Eqs. 6–9). At evaluation time, we run the recurrence over the observed history $\mathbf{S}^u = [s_1^u, \dots, s_{n_u}^u]$ and read out $\mathbf{y}_{\text{mem}}[L']$ at the final history position; the held-out test item $s_{n_u+1}^u$ is never fed into the memory update, preventing any leakage of the ground-truth label into the prediction.

3.4 Two-Phase Training Protocol

Joint training of encoder and memory risks competing gradient dynamics; we decouple them into two stages.

Phase 1 (Epochs 1–200): Base encoder training. We train the enhanced encoder (RoPE + KromHC + SwiGLU) on sequences of length at most $L = 300$, with TITANS entirely absent. The final checkpoint θ_{200} is retained as the frozen base initialization for Phase 2.

Phase 2 (Epochs 201–280): Memory module integration. We initialize the base encoder with θ_{200} , freeze all base encoder parameters, and train only the TITANS memory parameters ϕ_{mem} on extended sequences up to $L' = 600$, with early stopping (patience = 10 epochs on validation NDCG@10):

$$\min_{\phi_{\text{mem}}} \mathbb{E}_{(S, y)} [\mathcal{L}_{\text{rec}}(f_{\theta_{200}, \phi_{\text{mem}}}(S), y)]. \quad (11)$$

In practice, early stopping triggers after approximately 80 epochs on ML-1M and 60–70 epochs on sparser datasets. Because the base encoder is frozen, the attention-based prediction quality on short sequences is *structurally controlled*: $\mathbf{h}_{\text{attn}}[L]$ is identical to its Phase 1 value for any input, and the only modification to final scores is the additive term $\beta \cdot \mathbf{y}_{\text{mem}}[L']$. Note that β is part of ϕ_{mem} and is learned

to optimally weight the memory contribution; for users with short histories, y_{mem} carries limited useful signal (high associative error, few updates), and gradient descent trains β accordingly. This is not a hard zero-shot guarantee but a structural property: Phase 2 *cannot* degrade the base encoder’s short-sequence predictions it can only add the memory term, whose contribution is regulated by β .

3.5 Training Objective

The training loss is Generalized Binary Cross-Entropy (GBCE) [12]:

$$\mathcal{L}_{\text{GBCE}} = -\log \frac{\exp(\hat{y}_{\text{pos}}/\tau)}{\exp(\hat{y}_{\text{pos}}/\tau) + \sum_{i \in \mathcal{N}} \exp(\hat{y}_i/\tau)^\alpha}, \quad (12)$$

where $\mathcal{N}$ is a set of $K=100$ uniformly sampled negatives and $\alpha=0.75$ down-weights hard negatives to reduce representation degeneration. Optimization uses AdamW [19] with cosine annealing and 5% linear warmup.

3.6 Custom CUDA Kernel Implementations

The four CUDA kernels serve a dual purpose: they reduce per-iteration latency (enabling practical 300-epoch training), and critically for the TITANS kernel they make extended-sequence training memory-feasible at all. CONGA compiles all four extensions via `setuptools/ninja` against PyTorch’s C++ extension API. All kernels support float32 and float16 via PyTorch’s scalar-type dispatch, and share a common design discipline: fused forward-and-backward passes, closed-form gradient derivations, and warp-aligned thread layouts.

3.6.1 RoPE Kernel. The PyTorch eager path computes query and key rotations in two separate passes, each writing a full (B, L, H, D) tensor to global memory a total of $4BLHd$ bytes of round-trip DRAM traffic. Our kernel launches a 2-D grid of shape $(L, B \times H)$ that rotates *both* q and k in a single memory sweep using precomputed $(\cos, \sin)$ buffers:

$$(x_1, x_2) \mapsto (x_1 c - x_2 s, x_2 c + x_1 s), \quad (13)$$

reducing DRAM traffic to $2BLHd$ bytes ($2\times$ reduction) and eliminating the $O(BLHd)$ backward activation buffer entirely via analytic transpose rotation.

3.6.2 SwiGLU Kernel. Our kernel fuses the three-pass SwiGLU computation into a single 1-D flat grid of 256-thread blocks:

$$\text{out}[i] = \frac{w_1[i]}{1 + e^{-w_1[i]}} \cdot w_2[i], \quad (14)$$

reducing intermediate memory from $O(3n)$ to $O(n)$, with closed-form backward gradients derived analytically in the same pass.

3.6.3 KromHC Kernel. The kernel replaces iterative Sinkhorn-Knopp normalization $O(k \cdot M^2)$ for k iterations with the KromHC closed-form construction [8]:

$$\mathbf{H}_{\text{res}} = \mathbf{M}_1(a) \otimes \mathbf{M}_2(b), \quad \mathbf{M}_k(x) = \begin{bmatrix} x & 1-x \\ 1-x & x \end{bmatrix}, \quad a, b \in [0, 1], \quad (15)$$

producing an exact 4×4 doubly-stochastic matrix in $O(1)$ from two scalars. For $M=4$ and $k=20$ Sinkhorn iterations this yields a $5\times$ normalization cost reduction per block.

3.6.4 TITANS Kernel. A naïve autograd implementation of the TITANS recurrence must store the full $\mathbf{M}_t$ trajectory, incurring an $O(BLd^2)$ activation tape ≈ 1.17 GB at $B=128, L=600, d=64$, exceeding GPU memory. We resolve this via two co-designed optimizations. *Register tiling*: one CUDA block handles one batch item, and thread i owns row i of $\mathbf{M} \in \mathbb{R}^{d \times d}$ in registers throughout the recurrence, eliminating global-memory round-trips. *Chunked re-materialization* ($C=16$): the backward replays the forward recurrence chunk-by-chunk, reducing the activation footprint from $O(BLd^2)$ to $O(BCd^2)$ a $37.5\times$ reduction while maintaining exact gradients. Combined, these yield $13.2\times$ forward and $15.0\times$ backward speedups (Table 3) and make Phase 2 training at $L'=600$ feasible within 12 GB VRAM.

4 Experiments

We structure evaluation around four research questions:

- RQ1:** How does CONGA compare to state-of-the-art baselines under full-ranking evaluation?
- RQ2:** What is the individual contribution of each architectural component?
- RQ3:** Does CONGA maintain catalog diversity and avoid popularity bias?
- RQ4:** Does TITANS neural memory improve recommendation quality, and where does its benefit concentrate?

4.1 Datasets

We evaluate on four widely-used benchmarks. Dataset statistics are summarized in Table 2.

Table 2: Dataset statistics after preprocessing. Items with fewer than five interactions are excluded from evaluation; the reported item count reflects the evaluation catalog.

Stat	Beauty	Yelp	Steam	ML-1M
# Users	22,363	30,431	88,310	6,040
# Items	12,101	20,033	3,581	3,416
# Inter.	198,502	316,354	1,251,751	999,611
Avg. seq.	8.9	10.4	14.2	165.6
Sparsity	99.93%	99.95%	99.60%	95.20%

We adopt leave-one-out evaluation: the last interaction is the test item, the penultimate is validation, and all prior interactions are used for training. All four datasets are evaluated under a rigorous **full-ranking** protocol, scoring each test item against the complete item catalog without negative sampling, following the recommendation of [25].

4.2 Baselines

We compare against eight sequential recommendation baselines: **Caser** [17] (convolutional), **GRU4Rec** [10] (recurrent), **SASRec** [1] (causal self-attention, absolute PE), **BERT4Rec** [2] (bidirectional masking), **FMLPRec** [22] (frequency-domain MLP), **DuoRec** [12] (debaised contrastive), **FEARec** [23] (frequency-enhanced attention), **BSARec** [24] (behavior-sequence alignment), and **WEARec** [21] (wavelet-enhanced adaptive frequency filter; arXiv preprint, Nov. 2025) All models use $d=64$, 2 transformer blocks, 4 attention heads, dropout 0.2, 300 epochs, batch 128, lr 10^{-3} , weight decay 10^{-2} .

4.3 Metrics

Accuracy: $\text{HR}@K$ and $\text{NDCG}@K$ for $K \in \{5, 10\}$, evaluated under the full-ranking (all-item) protocol [25], which scores the ground-truth item against the *complete* catalog rather than a sampled subset. We adopt full ranking exclusively; sampled metrics are known to produce rankings inconsistent with full-catalog metrics [25] and can inflate or reverse relative model comparisons.

Diversity & robustness: Coverage@ K (fraction of catalog ever recommended), Head Bias (popularity concentration at the top-1 item), and Repetition Rate (fraction of already-consumed items in the top- K list, measuring interaction-masking correctness). Note that Repetition Rate is *distinct* from “hallucination” as used in generative settings: it measures whether the model leaks ground-truth history into predictions, not whether it fabricates non-existent items.

4.4 Implementation Details

All experiments use an NVIDIA RTX 5090 (32 GB) with CUDA 12.1 and PyTorch 2.9.1. Four custom CUDA kernels (RoPE, KromHC, SwiGLU, TITANS) are compiled via `setuptools/ninja`. Training uses `float32` precision. Code is available at: <https://anonymous.4open.science/r/RecSys-212/>.

4.5 CUDA Kernel Microbenchmarks

Table 3 reports per-kernel latency versus the equivalent PyTorch eager implementation. Measurements use `torch.cuda.Event` timing averaged over 100 warm-up + 500 timed iterations at $B=128$, $L=300$, `float32`, on an NVIDIA RTX 5090.

Table 3: Per-kernel forward and backward latency (μs): custom CUDA vs. PyTorch eager. Speedup = PyTorch / CUDA. Measurements at $B=128$, $L=300$, `float32`.

Kernel	Forward (μs)			Backward (μs)		
	CUDA	PyTorch	Speedup	CUDA	PyTorch	Speedup
RoPE	8.5	42.1	5.0×	12.2	65.4	5.4×
SwiGLU	15.2	58.6	3.9×	25.8	90.1	3.5×
KromHC	35.4	185.0	5.2×	50.1	240.5	4.8×
TITANS	95.0	1250.2	13.2×	140.5	2100.8	15.0×
Total	154.1	1535.9	10.0×	228.6	2496.8	10.9×

The **TITANS kernel** dominates (13.2× fwd, 15.0× bwd) via register tiling and chunked re-materialization ($C=16$), reducing the activation footprint from $O(BLd^2)$ to $O(BCd^2)$ and making Phase 2 at $L'=600$ feasible within 12 GB VRAM. The RoPE (5.0×), KromHC (5.2×), and SwiGLU (3.9×) kernels fuse multi-pass operations into single memory sweeps; combined, the 10.0× total speedup makes the full 300-epoch run feasible in under 5 hours.

5 Results

5.1 Overall Performance (RQ1)

Table 4 reports results across four benchmarks under full-ranking evaluation. Gains scale with interaction density: all three components compound on long-history datasets, while the RoPE backbone is the dominant driver on short-history ones (Section 6.1).

ML-1M ($\bar{L}=165.6$) is the only dataset where all three failure modes are simultaneously active. CONGA achieves +18.72% $\text{HR}@5$ and +18.77% $\text{NDCG}@10$ over WEARec, with RoPE closing the positional OOD gap, KromHC capturing orthogonal behavioral facets, and TITANS extending the effective context from 300 to 600 items.

Yelp ($\bar{L}=10.4$) yields gains of +11.99% $\text{HR}@5$ and +13.35% $\text{NDCG}@10$ over WEARec. Despite short mean sequences, high local interaction density provides sufficient signal for TITANS to contribute meaningfully, consistent with the +5.2% $\text{NDCG}@10$ gain at $A2 \rightarrow A3$ in ablation (Table 5).

Steam ($\bar{L}=14.2$) and **Beauty** ($\bar{L}=8.9$) gains of +3.32%/+6.16% $\text{HR}@5/10$ and +1.56% $\text{HR}@5$ respectively are driven primarily by the RoPE backbone; KromHC and TITANS do not contribute positively at these sequence lengths. On Beauty, FEARec retains an edge on $\text{HR}@10$ and $\text{NDCG}@10$ (−1.73%, −1.16%); practitioners on sparse corpora should deploy the backbone-only variant (A0).

Across the full 16-metric evaluation, CONGA ranks first on 14 out of 16 metrics. The dominant driver varies by regime: on long-history datasets (ML-1M, Yelp) all three components contribute cumulatively; on short-history datasets (Beauty, Steam) competitive performance reflects the strength of positional modernization, with the practical recommendation to deploy the backbone-only variant (A0) on sparse corpora.

5.2 Ablation Studies (RQ2)

Table 5: Incremental ablation on Yelp (sparse, $\bar{L}=10.4$) and ML-1M (dense, $\bar{L}=165.6$). Each row adds one component to the previous. Bold: best per column. $\uparrow/\downarrow$: gain/loss vs. previous row.

ID	Configuration	H@5	N@5	H@10	N@10
<i>Yelp (sparse, $\bar{L}=10.4$)</i>					
A0	Base encoder	0.0554	0.0365	0.0861	0.0463
A1	A0 + KromHC	0.0524 $\downarrow$	0.0339 $\downarrow$	0.0815 $\downarrow$	0.0433 $\downarrow$
A2	A1 + SwiGLU	0.0534 $\uparrow$	0.0350 $\uparrow$	0.0829 $\uparrow$	0.0444 $\uparrow$
A3	A2 + TITANS (full)	0.0551 $\uparrow$	0.0355 $\uparrow$	0.0901 $\uparrow$	0.0467 $\uparrow$
<i>ML-1M (dense, $\bar{L}=165.6$)</i>					
A0	Base encoder	0.2434	0.1675	0.3392	0.1984
A1	A0 + KromHC	0.2425 $\downarrow$	0.1669 $\downarrow$	0.3411 $\uparrow$	0.1988 $\uparrow$
A2	A1 + SwiGLU	0.2432 $\downarrow$	0.1698 $\uparrow$	0.3389 $\downarrow$	0.2007 $\uparrow$
A3	A2 + TITANS (full)	0.2435 $\uparrow$	0.1700 $\uparrow$	0.3402 $\uparrow$	0.2012 $\uparrow$

Baseline definition. Row A0 is a modernized SASRec encoder in which absolute positional embeddings are replaced by RoPE while all other design choices—causal self-attention, single-stream residual connection, GELU feed-forward network—remain unchanged. It is *not* the published SASRec baseline; rather, it constitutes a clean reference architecture that isolates the contributions of the three proposed components (KromHC, SwiGLU, TITANS) from the positional OOD confound that would otherwise inflate their apparent gains.

Including RoPE in A0 is motivated by a targeted control experiment on ML-1M: replacing only the positional encoding (RoPE vs.

Table 4: Overall performance comparison across four benchmarks under full-ranking (all-item) evaluation. Bold: best result. Underline: second best. Δ Improvement: relative gain of CONGA over the strongest baseline.

Dataset	Metric	Caser [WSDM'18]	GRU4Rec [ICLR'16]	SASRec [ICDM'18]	BERT4Rec [CIKM'19]	FMLPRec [WWW'22]	DuoRec [WSDM'22]	FEARec [SIGIR'23]	BSARec [AAAI'24]	WeaRec [arXiv'25]	CONGA	Improv.
Beauty	HR@5	0.0125	0.0169	0.0340	0.0469	0.0346	<u>0.0707</u>	0.0706	0.0694	0.0691	0.0718	1.56%
	HR@10	0.0225	0.0304	0.0531	0.0705	0.0559	0.0965	0.0982	0.0971	0.0948	<u>0.0965</u>	-1.73%
	NDCG@5	0.0076	0.0104	0.0221	0.0311	0.0222	0.0501	<u>0.0512</u>	0.0495	0.0490	0.0515	0.59%
	NDCG@10	0.0108	0.0147	0.0283	0.0387	0.0291	0.0584	0.0601	0.0585	0.0573	<u>0.0594</u>	-1.16%
Yelp	HR@5	0.0117	0.0130	0.0149	0.0256	0.0159	0.0271	0.0262	0.0453	<u>0.0492</u>	0.0551	11.99%
	HR@10	0.0197	0.0221	0.0249	0.0433	0.0287	0.0442	0.0437	0.0724	<u>0.0781</u>	0.0901	15.36%
	NDCG@5	0.0070	0.0080	0.0091	0.0159	0.0100	0.0170	0.0165	0.0293	<u>0.0320</u>	0.0355	10.94%
	NDCG@10	0.0096	0.0109	0.0123	0.0216	0.0142	0.0225	0.0221	0.0380	<u>0.0412</u>	0.0467	13.35%
ML-1M	HR@5	0.0927	0.1005	0.1374	0.1512	0.1316	0.1838	0.1834	0.1944	<u>0.2051</u>	0.2435	18.72%
	HR@10	0.1556	0.1657	0.2137	0.2346	0.2065	0.2704	0.2705	0.2757	<u>0.2987</u>	0.3402	13.89%
	NDCG@5	0.0592	0.0619	0.0873	0.1021	0.0846	0.1252	0.1236	0.1306	<u>0.1393</u>	0.1700	22.04%
	NDCG@10	0.0795	0.0828	0.1116	0.1289	0.1087	0.1530	0.1516	0.1568	<u>0.1694</u>	0.2012	18.77%
Steam	HR@5	0.0304	0.0606	0.0683	0.0645	0.0464	<u>0.0754</u>	0.0701	0.0694	0.0700	0.0779	3.32%
	HR@10	0.0533	0.1021	0.1105	0.1054	0.0772	<u>0.1201</u>	0.1108	0.1101	0.1101	0.1275	6.16%
	NDCG@5	0.0190	0.0380	0.0431	0.0417	0.0300	<u>0.0503</u>	0.0455	0.0455	0.0459	0.0513	1.99%
	NDCG@10	0.0263	0.0514	0.0567	0.0548	0.0398	<u>0.0646</u>	0.0584	0.0585	0.0588	0.0672	4.02%

absolute PE, all other hyperparameters fixed) yields +3.9% NDCG@10, confirming that positional OOD degradation is an independent, dominant factor on long-history data. Absorbing this fix into the baseline ensures that rows A1–A3 strictly attribute incremental gains to KromHC multi-stream fusion, SwiGLU, and TITANS memory, respectively, rather than to positional encoding choice.

Ablation Datasets. We evaluate the ablations on **Yelp** and **ML-1M** as two representative endpoints. Following the convention in prior work [1, 24], we select subsets most diagnostic for the proposed mechanisms: Yelp represents the **sparse, short-sequence** regime ($\bar{L}=10.4$, sparsity 99.95%), while ML-1M captures the **dense, long-sequence** regime ($\bar{L}=165.6$, sparsity 95.20%). Steam is intentionally omitted from this study, as its interaction pattern closely mirrors the ML-1M dense regime at a smaller scale, yielding non-discriminative ablation deltas.

Table 5 reveals consistent but *dataset-dependent* contribution patterns.

On ML-1M, KromHC provides a marginal gain (A0→A1: +0.2% NDCG@10) the Kronecker mixing matrix converges before attention weights fully stabilize under the fixed training budget. SwiGLU resolves this gradient-flow timing issue (A1→A2: +0.95% NDCG@10), and TITANS adds +0.25% (A2→A3), primarily structural (zero repetition rate, forgetting prevention) given the ≈ 80 -epoch Phase 2 budget. A3 strictly outperforms A0 on all four metrics, with a combined gain of +1.41% NDCG@10 from KromHC, SwiGLU, and TITANS on top of the RoPE-equipped backbone.

On Yelp, KromHC introduces a substantial dip (A0→A1: −6.5% NDCG@10, −5.4% HR@5) at $\bar{L}=10.4$, right at the boundary of the adaptive $M=4$ rule (Eq. 3), the additional stream capacity over-parameterizes the sparse signal before attention weights have converged. SwiGLU partially recovers (A1→A2: +2.5% NDCG@10) and TITANS provides the largest individual gain (A2→A3: +5.2% NDCG@10). The final model A3 achieves the best HR@10 and NDCG@10 across all rows; HR@5 and NDCG@5 remain marginally

below A0 (0.0551 vs. 0.0554; 0.0355 vs. 0.0365), reflecting a precision-recall trade-off: TITANS broadens associative recall across longer list positions at the cost of marginal top-5 precision.

Despite non-monotone per-step trajectories on both datasets, A3 strictly outperforms A0 on all metrics for ML-1M and achieves the best HR@10 and NDCG@10 on Yelp, confirming collective complementarity over redundancy. The adaptive stream-selection rule (Eq. 3) and learned β handle regime-specific calibration automatically, so the full CONGA stack requires no dataset-specific tuning.

5.3 Robustness and Diversity (RQ3)

We report five complementary metrics alongside accuracy:

- (i) **Catalog Coverage@K** measures the fraction of the item catalog ever recommended:

$$\text{Coverage@K} = \frac{|\bigcup_{u \in \mathcal{U}} \text{Rec}_K(u)|}{|\mathcal{I}|}. \quad (16)$$

- (ii) **ILD** (Intra-List Diversity): average pairwise cosine distance within each user’s top- K list; higher values indicate more diverse recommendations.
- (iii) **Tail Ratio**: fraction of recommended slots occupied by bottom-20% popularity items; higher values indicate less head-bias toward popular items.
- (iv) **Head Bias** measures the fraction of users who receive the single most-recommended item:

$$\text{HeadBias} = \max_{i \in \mathcal{I}} |\{u : i \in \text{Rec}_K(u)\}| / |\mathcal{U}|, \quad (17)$$

where a lower value indicates less popularity concentration.

- (v) **Repetition Rate** measures the fraction of recommended slots occupied by items the user has already consumed:

$$\text{RepRate} = \frac{\sum_u |\text{Rec}_K(u) \cap \mathcal{H}_u|}{|\mathcal{U}| \cdot K}, \quad (18)$$

where $\mathcal{H}_u$ is user u 's training history; a non-zero value indicates a masking defect.

Table 6: Diversity and robustness metrics on ML-1M (top-10). ILD = Intra-List Diversity (avg. pairwise cosine distance; higher = more diverse). Tail Ratio = fraction of recommended slots from bottom-20% popularity items. Head Bias = fraction of users receiving the single most-popular recommended item (lower = less concentrated). Rep. Rate = fraction of already-consumed items in top- K (non-zero = masking defect).

Metric	CONGA-base	CONGA-full	Δ
Coverage@10	74.68%	74.80%	+0.12 pp
ILD	0.6528	0.6235	−0.029
Tail Ratio	0.94%	0.77%	−0.17 pp
Head Bias	5.58%	6.36%	+0.78 pp
Rep. Rate	0.00%	0.00%	±0

Coverage increases marginally (+0.12 pp) as TITANS broadens associative recall across extended histories. ILD and Tail Ratio decline slightly (−0.029 and −0.17 pp respectively), and Head Bias rises by +0.78 pp a classical accuracy–diversity trade-off: the memory module reinforces popular co-occurrence patterns extracted from long interaction histories, marginally concentrating recommendations toward high-affinity items. These shifts are small in absolute terms and consistent with the accuracy gains reported in Table 4; they do not indicate a structural bias introduced by the TITANS module. Repetition Rate is exactly 0.00% for both variants, confirming correct interaction-masking across all $6,040 \times 10$ recommendation slots.

5.4 Long-History User Analysis (RQ4)

To validate that TITANS neural memory disproportionately benefits users with richer interaction histories consistent with the two-phase training rationale we partition the 6,040 ML-1M users into three groups by total interaction count (training + validation history), and report NDCG@10 per group for CONGA-base (no TITANS) and Full CONGA.

Table 7: NDCG@10 by user history length on ML-1M ($|\mathcal{U}| = 6,040$). Users are partitioned by total interaction count. $\Delta\%$: relative gain of Full CONGA over CONGA without TITANS.

Group	Users	w/o TITANS	CONGA-full	$\Delta\%$
Short (≤ 50 items)	1,837	0.2535	0.2536	+0.04%
Medium (51–150 items)	2,130	0.2024	0.2027	+0.15%
Long (> 150 items)	2,073	0.1521	0.1532	+0.72%
Overall	6,040	0.2007	0.2012	+0.25%

Table 7 shows that the relative improvement grows *monotonically* with interaction history length: +0.04% for short-history users (≤ 50 interactions), +0.15% for medium, and +0.72% for long-history users (> 150 interactions). This monotone pattern directly validates the TITANS integration rationale: the $O(\sqrt{T})$ regret bound of the online

update rule predicts that longer histories supply more gradient updates to $\mathbf{M}_t$, enabling increasingly reliable associative recall. The learned fusion scalar $\beta \in \phi_{\text{mem}}$ provides a complementary self-regulating mechanism converging toward zero for short-history users where associative error is high and the memory carries limited signal, and toward larger values where history depth justifies a meaningful memory contribution. This self-regulation ensures that Phase 2 training does not degrade base-encoder quality on sparse histories: the +0.04% gain on short-history users confirms that the frozen-encoder design successfully isolates the memory term from the learned attention representations.

6 Discussion

6.1 Regime-Dependent Behaviour

Each component's benefit is conditional on corpus interaction density. **RoPE** generalizes unconditionally, contributing positively across all four benchmarks via its norm-preserving property. **KromHC** requires sufficient sequence diversity: on short-sequence corpora (Beauty, $\bar{L}=8.9$; Steam, $\bar{L}=14.2$), the mixing matrix memorizes spurious co-occurrence patterns before attention weights converge, producing a dip at $A_0 \rightarrow A_1$ even at $M=2$; the adaptive threshold (Eq. 3) mitigates but does not fully resolve this for $\bar{L} \in [10, 50]$. **TITANS** similarly degrades on Steam (−11.3% NDCG@5 at $A_2 \rightarrow A_3$), consistent with its $O(\sqrt{T})$ regret bound; the learned β partially self-regulates but does not fully compensate within the 80-epoch Phase 2 budget. **CONGA is therefore optimized for $\bar{L} \gtrsim 50$** ; on sparse corpora the backbone-only variant (A_0) is preferred.

6.2 Limitations and Future Work

Short-history regimes. KromHC and TITANS contribute negatively for $\bar{L} \leq 50$. A learned adaptive gate replacing the fixed rule in Eq. 3 is a natural extension. **Cold-start and modality.** Users with fewer than ~ 10 interactions receive no memory benefit; integrating textual or visual features would enrich the associative key-value representations and alleviate cold-start. **Memory scalability.** The $O(d^2)$ -per-user TITANS footprint restricts batch size at large d ; low-rank factorization or cross-user shared memory are promising remedies.

7 Conclusion

We presented CONGA, which transfers RoPE, KromHC, SwiGLU, and TITANS neural memory into sequential recommendation via a two-phase forgetting-prevention protocol. The three contributions—failure-mode diagnosis, first TITANS adaptation to discrete-item recommendation, and regime-dependent empirical validation collectively yield state-of-the-art accuracy on long-history benchmarks (+18.72% HR@5 on ML-1M, +13.35% NDCG@10 on Yelp) with zero repetition rate across all configurations. On short-history corpora the RoPE backbone alone is competitive, identifying a clear architectural scope boundary. Promising extensions include cross-domain memory transfer, continual catalog adaptation, and multi-modal item features. We hope CONGA establishes a replicable blueprint for closing the architecture gap between language modelling and recommender systems—demonstrating that principled transfer of LLM advances, rather than wholesale adoption, is both tractable and effective.

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